INCH POUND

MIL-C-3098/146A(ER)
23 December 1991

SUPERSEDING
MIL-C-3098/146(ER)
24 September 1986

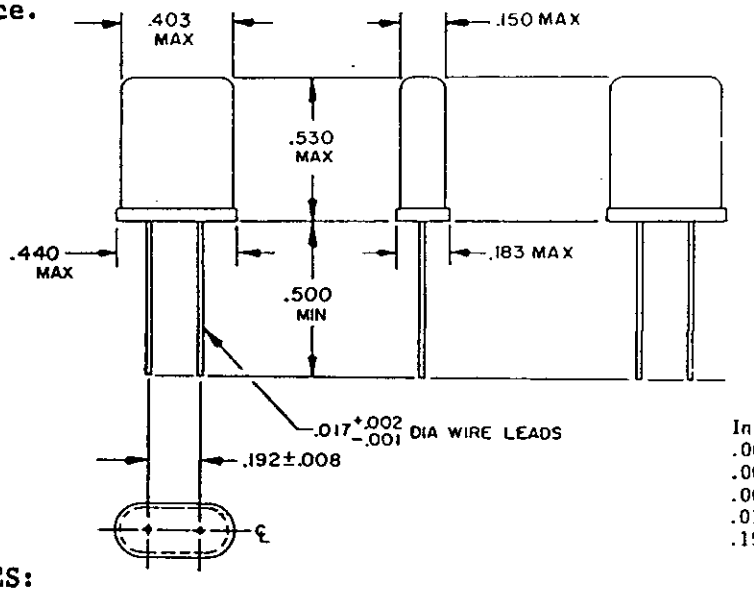
MILITARY SPECIFICATION SHEET

CRYSTAL UNIT, QUARTZ, CR165/U

This specification is approved for use by the Depart ment of the Army and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-3098.

Pertinent characteristics: 2.4 to 20 MHz; fundamental; noncontrolled; antiresonance.



Inches	mm	Inches	mm
.001	0.03	.183	4.65
.002	0.05	.192	4.88
.008	0.20	.403	10.24
.017	0.43	.435	11.05
.150	3.81	.500	12.70
		.530	13.46

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Corrosion resistance metal plating of the enclosure after resistance welding is permitted.
4. Marking shall be in accordance with MIL-C-3098.

FIGURE 1. Crystal unit - CR165/U.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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REQUIREMENTS:

Holder: HC-49/U (MIL-H-10056/36).

Dimensions, marking, and configuration: See figure 1.

Frequency range: 2.4 to 20 MHz, inclusive.

Frequency tolerance: ±50 ppm.

Equivalent resistance: See TABLE I.

TABLE I Equivalent resistance.

Frequency range, inclusive. MHz.	Maximum resistance Ohms
2.4 to 2.60	300
2.61+ to 2.89	250
2.9 to 3.75	180
3.75+ to 4.75	120
4.75+ to 6.0	75
6.0+ to 7.0	50
7.0+ to 10.0	30
10.0+ to 20.0	25

Mode of oscillation: Fundamental.

Operating temperature range: -55°C to +105°C, inclusive.

Drive level: 5.0 ±1.0 mW.

Antiresonance, load capacitance: 30.0 ±0.5 pF.

Capacitance, shunt:
2.4 MHz to 4.0 MHz --- 1.7 pF min to 7.0 pF max.
4.0 MHz to 6.0 MHz --- 2.7 pF min to 7.0 pF max.
6.0 MHz to 20.0 MHz --- 3.7 pF min to 7.0 pF max.

Shock:
Frequency change permitted: ±5 ppm.
Equivalent resistance change permitted: ±10 percent.

Vibration: Method 201 Of MIL-STD-202, test condition A.
Frequency change permitted: ±5 ppm.
Equivalent resistance change permitted: ±10 percent.

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Temperature cycling:
  Frequency change permitted: ±5 ppm.
  Equivalent resistance change permitted: ±10 percent.

Aging:
  Frequency change permitted: ±5 ppm.

Revision letters are not used to denote changes due to the extensiveness of the changes.

CONCLUDING MATERIAL

Preparing activity:
  Army-ER

Agent:
  DLA - ES

(Project 5955-A207)

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